



Refined Insight into the Wingtip Vortex-Free Shear Layer Interaction

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Abstract

Interaction between the free shear layer (FSL), the wingtip vortex and aerodynamic efficiency was observed. These results were quantitatively evaluated through Particle Image Velocimetry (PIV) of the wingtip vortex and the free shear layer (FSL) obtained in the Low Speed Wind Tunnel at the University of Dayton (UD-LSWT). The experiments were conducted in the wake of an AR 4 flat plate with and without a boundary layer trip (BLT). The effect of the BLT on both the FSL and the wingtip vortex was previously documented. In the present work, the analysis was extended to explore the role the FSL interaction plays in the wingtip vortex evolution. Additionally, a correlation between this FSL-wingtip vortex interaction and the aerodynamic efficiency of the wing was sought. Streamwise, cross-stream and spanwise plane oriented PIV of the wingtip vortex shows evidence of the FSL interaction with the wingtip vortex at lower angles of attack. Due to the FSL interaction with the wingtip vortex, vortex profiles predicted using Batchelor's and Brown's models deviate from the experimental data. The composite of the experimentally determined velocity profiles from different planes implied an exchange of momentum between the FSL and the wingtip vortex in the vicinity of the maximum (L/D) lift condition. At an angle of attack less than that of maximum (L/D), the additional loss of momentum due to the BLT is seen in the wing wake FSL. At an angle of attack greater than that corresponding to the maximum (L/D), the additional loss of momentum due to the BLT is instead seen in the wingtip vortex.

1. Introduction

The prevalence of wingtip devices (winglets, blended winglets, etc.) on commercial airliners evidences the continued importance of lift induced drag reduction on modern aircraft. From a global perspective the following study has been performed to further explore the wingtip vortex evolution with the underlying motivation that a more elegant solution to the reduction of lift induced drag must exist. This motivation was the inspiration for the original work by Gunasekaran and Altman [1].

The present study is a follow-on to [1] intended to explore the interaction between the FSL and the wingtip vortex and their collective correlation to the maximum (L/D) lift condition where the authors explored the effect on the free shear layer (FSL) and the wingtip vortex due to the placement of a spanwise boundary layer trip (BLT) on the upper surface of an AR 4 flat plate. The Particle Image Velocimetry (PIV) results indicated significant changes in mean and fluctuating velocities in both the FSL and in the wingtip vortex due to the BLT as hypothesized. This hypothesis was based on Brown's model [2] which assumes that profile drag of the wing is enveloped in the roll-up process of wingtip vortex. The significant changes in the axial and the azimuthal velocity of the wingtip vortex documented in [1] provided evidence in support of Brown's underlying assumption with respect to the vortex rollup process. The changes in the FSL streamwise velocity, wingtip vortex axial velocity and wingtip vortex azimuthal velocity indicated there might be an interaction among them. Assuming the properties in the wingtip vortex act as a surrogate for induced drag, and the properties in the FSL act as a surrogate for parasite drag, any indication of interaction between the FSL and the wingtip vortex will have serious implications on altering the aerodynamic

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efficiency of the wing. This paper will focus on detailing that physical interaction between the FSL and the wingtip vortex and will also indicate this interaction satisfies conservation of momentum in the wake of the wing. Three-dimensional conservation of momentum between the FSL and the wingtip vortex will be discussed in the future paper.

Marasli et al [3] showed that the properties in the FSL in the wake of an object upstream are unique to turbulent generators. Subsequently Gunasekaran and Altman [4] showed that not only is the FSL unique but the performance information of the wing is preserved in the FSL at relatively large distances downstream of the wing. Therefore, the inherent connection between the upstream flow over wing and the properties seen in the wake provides the fundamental motivation behind the present study. Currently, most commercial airplanes do not cruise at the lift condition associated with maximum aerodynamic efficiency because it is too slow. It is hypothesized that a better understanding of the relationship between the aerodynamic efficiency, the properties in the wingtip vortex and the properties in the FSL can be used to improve the aerodynamic efficiency at more conventional cruise conditions.

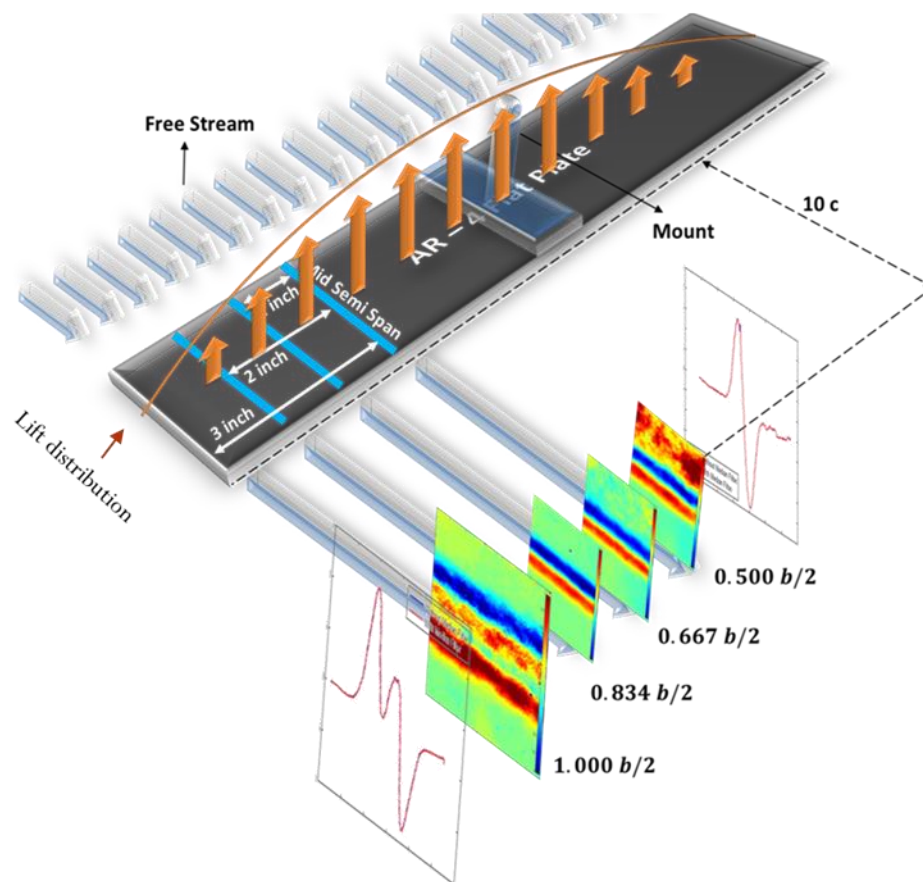


Figure 1 Variation of Reynolds stress at different spanwise stations (0.5 $b/2$, 0.667 $b/2$, 0.834 $b/2$ and 1.000 $b/2$ (Wingtip)). The wingtip vortex is essentially preserved 10 chord lengths downstream sandwiched between nearly identical Reynolds stress distributions from inboard wing stations [3].

The experimental evidence of this particular type of wingtip vortex-FSL interaction was observed in an experiment conducted in the Horizontal Free Surface Water Tunnel (HFWT) at the Air Force Research Labs (AFRL) on an AR 4 flat plate [3]. The experiments were done in support of wake identification research where the properties in the FSL wake are used to identify the turbulent generator upstream. The Reynolds stress contours of the FSL across different spanwise stations in the wake of an AR 4 flat plate at

a 2° angle of attack is shown in Figure 1 [3]. At this particular angle of attack, the wingtip vortex is seen to be sandwiched between the shear layers at the wingtip spanwise station. It is strange that despite the complex dynamics and comparatively violent mixing associated with the wingtip vortex rollup process, the shear dominated wake remains neatly stratified and its turbulence characteristics preserved on either side of the wingtip vortex, at least as far as 10 chord lengths downstream. This type of interaction between the wingtip vortex and the FSL has not been previously identified in the literature.

In the present study, this surprising interaction between the FSL and the wingtip vortex is studied in detail by expanding upon experimental investigations documented in [1]. A brief summary of the experimental setup is repeated in this paper for completeness. The PIV experiments discussed here are conducted in the Low Speed Wind Tunnel at the University of Dayton (UD-LSWT) at an approximate Reynolds number of 150,000. The on-body boundary layer, mid-semispan free shear layer and the evolution of the wingtip vortex of an AR 4 flat plate were examined using the Particle Image Velocimetry (PIV) technique at 2, 3, 4, and 5 chord lengths downstream of the trailing edge of the wing. The evolution of the wingtip vortex was interrogated in three different planes oriented along the three axes (Streamwise, Cross-stream and Spanwise). Then a boundary layer trip was introduced to change the character of the boundary layer and the nature of its associated turbulence over the upper surface of the flat plate (Figure 3). The resulting change influenced turbulence in the free shear layer and in the wingtip vortex as determined by analyzing the mean and the fluctuating quantities in the wake. The PIV planes of interrogation in the wake of the flat plate are shown in Figure 2. The test parameters of the experiment conducted are shown in Table 1.

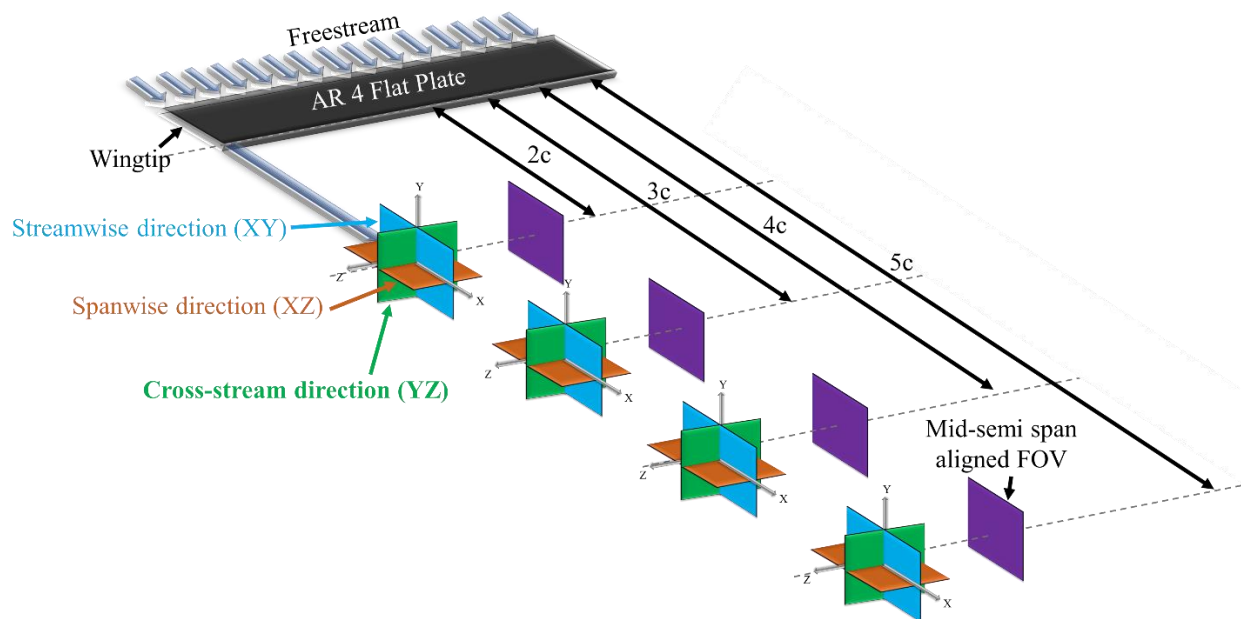


Figure 2 3D schematic of PIV Planes (Mid semispan, Streamwise (XZ), Cross-stream (YZ) and Spanwise (XY)) with respect to the flat plate.

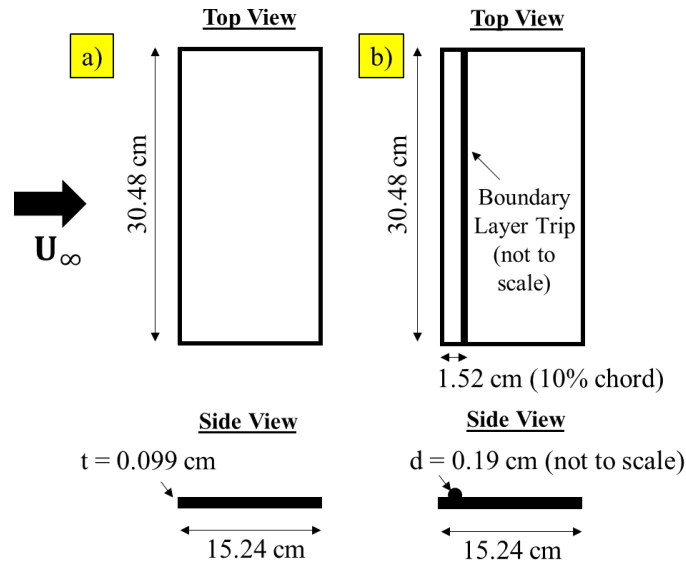


Figure 3 a) AR 4 flat plate without a BLT b) AR 4 flat plate with a spanwise BLT 0.19 cm in diameter placed at 10% chord.

Table 1 Specifications of the flat plate tested

Specifications of the Flat Plate Tested	
Aspect Ratio	4
Chord	15.24 cm
Half-Span	30.48 cm
t/c ratio	0.6%
Leading edge shape	Squared
Freestream Velocity	15 m/s
Reynolds Number	151698

2. Background: Theoretical Models

In order to quantify the effect of the BLT on the wingtip vortex, the experimental results were compared with three well-known wingtip vortex models available from the literature.

2.1 Batchelor's model

One of the original wingtip vortex models to include axial core velocity considerations was published by Batchelor [5] in 1964. Unlike the line vortex derived from elementary potential flows, there exists an axial velocity in the core of a wingtip vortex. Assuming axial gradients in the wingtip vortex are smaller when compared to the radial gradients in the wingtip vortex, Batchelor used the Equations of motion for steady axisymmetric flow of an incompressible fluid to obtain the relationship between the wingtip vortex core axial velocity and the azimuthal velocity (Equation 1).

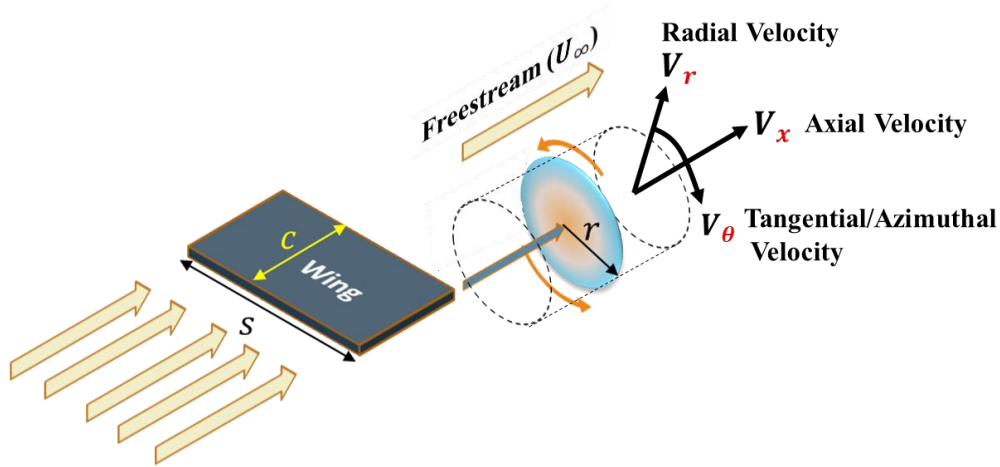


Figure 4 Wing-Wingtip vortex schematic with the coordinate axis definition (One tip vortex is shown for clarity). A cylindrical coordinate system is used for analysis with the control volume enveloping the wingtip vortex.

From Figure 4, assuming steady, incompressible flow, the conservation of radial momentum Equation can be represented as,

$$V_x \frac{\partial V_r}{\partial x} + V_r \frac{\partial V_r}{\partial r} - \frac{V_\theta^2}{r} = -\frac{1}{\rho} \frac{\partial P_{vortex}}{\partial r} + \nu \left(\frac{V_r}{r^2} - \nabla^2 V_r \right) \quad (1)$$

where ρ is density, P_{vortex} is the pressure in the wingtip vortex, V_x is the axial velocity, V_r is the radial velocity, V_θ is the wingtip vortex azimuthal velocity and r is the radius of the wingtip vortex. Assuming there is no radial velocity V_r in the wingtip vortex, Equation 1 simplifies to,

$$\frac{V_\theta^2}{r} = \frac{1}{\rho} \frac{\partial P_{vortex}}{\partial r} \quad (2)$$

Equation 2 is applicable for both viscous and inviscid assumptions since the magnitude and gradient of the radial velocity is negligible. Hence, even with the viscous assumption, the viscous force term on the right hand side of the Equation 1 can be neglected.

Continuing with the presentation of relevant equations, from potential flow theory (assuming inviscid, incompressible flow, steady flow), the tangential velocity can be represented as,

$$V_\theta = -\frac{\Gamma}{2\pi r} \quad (3)$$

Where r is the radius of the vortex and Γ is the circulation of the vortex. Substituting Equation 3 in Equation 2 and integrating from a radius r to ∞ ,

$$\int_r^\infty \frac{\partial P_{vortex}}{\rho} = \left(\frac{\Gamma}{2\pi} \right)^2 \int_r^\infty \frac{\partial r}{r^3} \quad (4)$$

Considering the pressure goes to P_0 at ∞ , Equation 4 can be written as,

$$\left(\frac{\Gamma}{2\pi} \right)^2 = \frac{2r^2(P_0 - P_{vortex})}{\rho} \quad (5)$$

From Equation 5, assuming the circulation and density of the wingtip vortex remains constant as a function of downstream distance, if the core diameter (r) increases with downstream distance, the pressure in the core (P_{vortex}) increases leading to axial deceleration (wake-like profile). If the core diameter decreases with the distance downstream, the pressure in the core decreases leading to axial acceleration (jet-like profile).

Even though Equation 1 was derived using an inviscid approach, Batchelor suggested that viscous dissipation will result in an axial total pressure loss that would not be negligible at a higher Reynolds number. He also theorized that viscous diffusion will ultimately lead to an increase in vortex core diameter with distance downstream as $x^{1/2}$ where x is the downstream distance. The gradual decrease in azimuthal velocity by viscous diffusion increases the pressure in the core of the vortex leading to an axial deceleration. He derived a similarity solution for the asymptotic decrease in the axial core velocity deficit with downstream distance (x) due to viscous diffusion as shown in Equation 6. The asymptotic decrease of normalized axial velocity with normalized downstream distance due to viscous dissipation is shown in Figure 5.

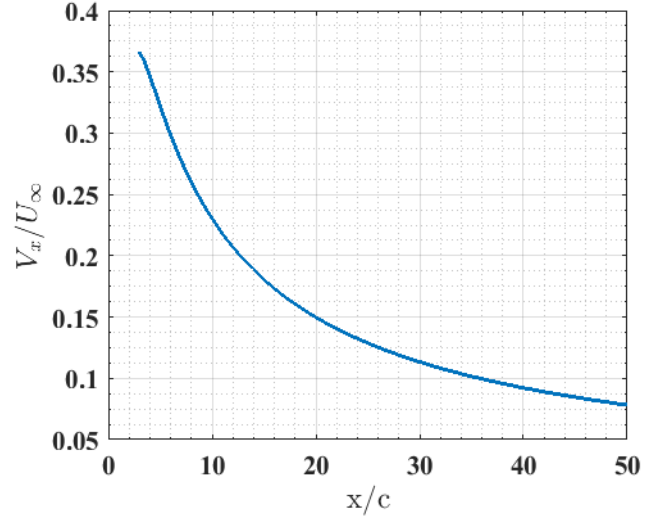


Figure 5 An asymptotic decrease of wingtip vortex core axial velocity with downstream distance due to viscous dissipation as theorized by Batchelor

$$\frac{V_x}{U_\infty} \propto \left(\frac{x}{c}\right)^{-1} \log\left(\frac{x}{c}\right) \quad (6)$$

A helpful analogy to Batchelor's model of the wingtip vortex is that of the vortex produced by stirring coffee in a coffee cup. As the azimuthal stir velocity increases, the vortex diameter decreases and the pressure in the vortex decreases leading it to descend into the coffee cup. As the stir velocity decreases, the vortex diameter increases and the pressure in the vortex increases leading it to ascend from the bottom of the cup. As demonstrated by the example, any change in the azimuthal velocity of the wingtip vortex will result in changes in the wingtip vortex core axial velocity. The azimuthal velocity and axial velocity component of the wingtip vortex are derived from the similarity solutions.

Batchelor's relation for the azimuthal velocity of the wingtip vortex:

$$\frac{V_\theta(\eta)}{V_{\theta_{max}}} = \left(1 + \frac{1}{2\alpha}\right) \frac{1}{\eta} (1 - \exp(-\alpha\eta^2)) \quad (7)$$

Batchelor's relation for the axial velocity of the wingtip vortex:

$$\frac{V_x(\eta)}{V_{x_{max}}} = \exp(-\alpha l^2 \eta^2) \quad (8)$$

where $V_\theta(\eta)$ is the azimuthal velocity, $V_{\theta_{max}}$ is the maximum azimuthal velocity, $\eta = r/r_c$ is the radial coordinate with r_c being the core radius, α is Lamb's constant with a value of 1.256, $V_x(\eta)$ is the axial velocity, $V_{x_{max}}$ is the peak axial velocity, l is a rescaling parameter for the axial profile.

2.2 Brown's Model

In 1973, Brown [2] extended Batchelor's model by relating the axial core velocity to the profile drag (drag due to pressure drag and skin friction drag on the wing) and the induced drag of the wing. Brown hypothesized that all of the profile drag is enveloped in the rollup of wingtip vortex and the nature of the core axial velocity in the wingtip vortex is a function of the balance between the induced drag and the profile drag of the wing. He formulated a mathematical model (Equation 9) for the axial core velocity of a wingtip vortex based on Batchelor's model for axial flow associated with a rotating flowfield. Brown's model for the wingtip vortex core axial velocity is,

$$\frac{\Delta V_x}{U_\infty} = \left[\frac{3C_L^2}{4\pi^4 AR^2} - \frac{3C_{D_{Profile}} c \left(\frac{r}{s}\right)}{8\pi AR \bar{c}} \right] \frac{s}{r} \quad (9)$$

where ΔV_x is the incremental core axial velocity in the wingtip vortex, U_∞ is the freestream velocity, C_L is the coefficient of lift of a finite wing, AR is the aspect ratio of the wing, $C_{D_{Profile}}$ is the profile drag coefficient, c is the local chord, $\bar{c}$ is the mean chord, s is the span and r is the radius of the wingtip vortex. The variables in Equation 9 are annotated in the wing-wingtip vortex schematic shown in Figure 4.

Assuming an elliptical lift distribution (span efficiency of 1) and a rectangular wing ($c = \bar{c}$), Equation 9 can be rewritten in terms of lift induced drag and profile drag as,

$$\frac{\Delta V_x}{U_\infty} = \frac{3}{4\pi AR} \left[\frac{C_{D_{Induced}}}{\pi^2} - \frac{C_{D_{Profile}}}{2} \right] (r/s) \quad (10)$$

From Equation 10, Brown determined that if

$$C_{D_{Induced}}/\pi^2 \geq C_{D_{Profile}}/2 \quad (11)$$

the wingtip vortex axial core velocity will be greater than the freestream velocity resulting in a jet-like behavior. Alternatively, if

$$C_{D_{Induced}}/\pi^2 \leq C_{D_{Profile}}/2 \quad (12)$$

the wingtip vortex axial core velocity will exhibit a wake-like behavior where the magnitude of the axial core velocity is less than the freestream velocity.

3. Results

4.1 Interaction Between the FSL Streamwise Velocity and the Wingtip Vortex Core Axial Velocity

The wingtip vortex streamwise PIV measurements were obtained at several discrete distances downstream of the flat plate with and without a BLT at 1°, 2°, 3°, and 4° angles of attack (Figure 2). The wingtip vortex axial velocity contours are shown in Figure 6 for 1°, 2°, 3° and 4° angle of attack at several distances downstream of the flat plate. The difference in the freestream velocity as a function of downstream distance is less than 1%. This slight decrease in the velocity is due to the expansion of the jet in the open jet configuration in the UD-LSWT.

At a 1° angle of attack, the wingtip vortex is feeble and the wake is shear layer dominated since a minuscule amount of lift is being produced by the flat plate at this angle of attack. The wingtip vortex and the shear layer are indistinguishable at this angle of attack (Figure 6). As downstream distance increases

(5c downstream), subtle differences can be seen between FSL and the wingtip vortex. At a 2° angle of attack, the flat plate generates sufficient lift for both the shear layer and the wingtip vortex to be observed. At this angle of attack, bifurcation of the wingtip vortex and the shear layer occurs but they are still difficult to discern independently. At a 3° angle of attack, a clear bifurcation of the wingtip vortex and shear layer can be extracted from the contours. As the angle of attack is increased to 4° , the shear layer convects downwards from the wingtip vortex and distinguishable wake signatures of the shear layer and the wingtip vortex are plainly observed. It is fascinating to observe that the wingtip vortex divorces from the shear layer in the vicinity of the maximum (L/D) angle of attack of the flat plate ($\sim 3^\circ$ angle of attack).

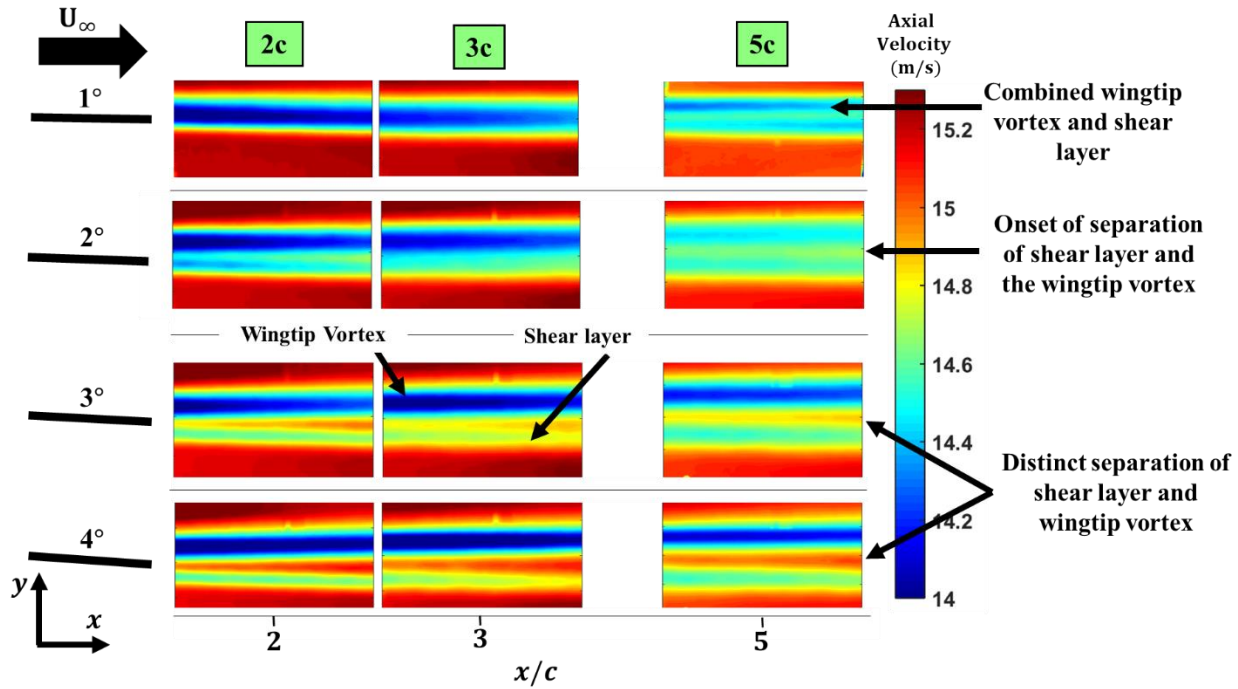


Figure 6 Axial/streamwise velocity contours of wake propagation downstream of a flat plate at the wingtip spanwise location at different angles of attack. The interaction of the shear layer and the wingtip vortex can be clearly seen at lower angles of attack. The shear layer separates from the wingtip vortex in the vicinity of the maximum (L/D) lift condition.

The comparison of the FSL normalized streamwise velocity profile and the wingtip vortex normalized axial velocity profile obtained from the UD-LSWT is shown in Figure 7 for different angles of attack. At 1° and 2° angles of attack, the peak normalized velocity of the mid semispan FSL is lower than the peak normalized axial core velocity of the wingtip vortex. At 3° and 4° angle of attack, this trend is reversed where the peak normalized wingtip vortex axial core velocity is lower than the peak normalized streamwise velocity of the mid semispan FSL. As the angle of attack increases, the peak normalized wingtip vortex core axial velocity decreases. This decrease in the wingtip vortex core axial velocity with increase in angle of attack will next be evaluated in the context of Brown's model (Equation 10).

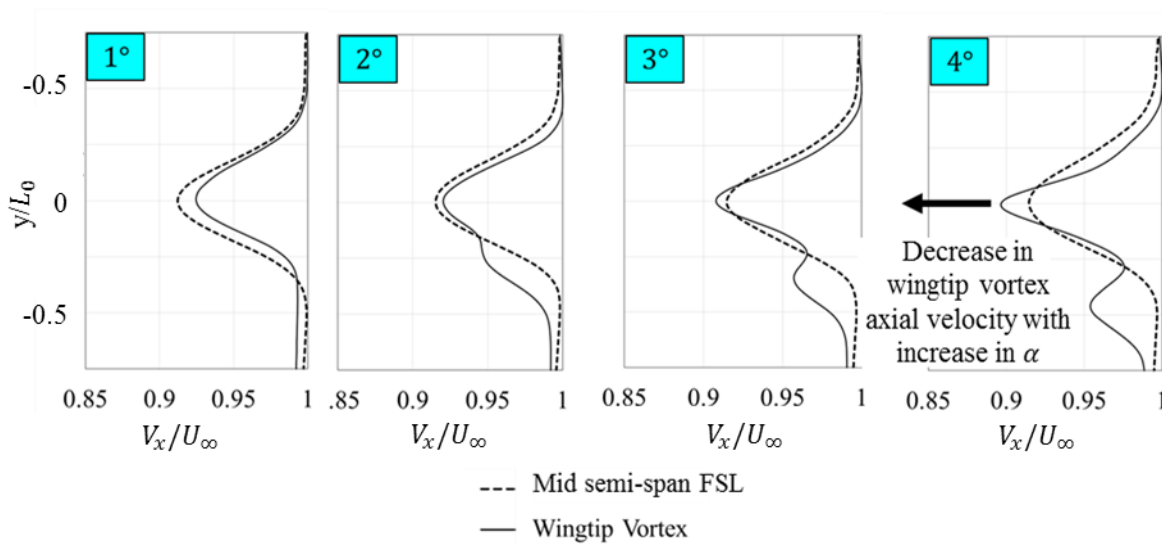


Figure 7 Comparison of peak normalized streamwise/axial velocity (V_x/U_∞) between the mid-semispan FSL and the wingtip vortex at 3 chord lengths downstream for the tripped case. The y-axis is normalized based on wake-half width L_0 . The peak normalized axial velocity of the wingtip vortex decreases more than the peak normalized velocity of the mid semispan FSL as the angle of attack increases. Also, you need to change your independent axis titles to reflect the fact that for the FSL, it is not an axial velocity.

The experimental coefficient of lift and drag results obtained from Mueller [7] were substituted into Brown's equation (Equation 10) and the axial velocity profiles were obtained. The result is shown in Figure 8. The increase in angle of attack results in a decrease in the peak normalized wingtip vortex axial core velocity of the wingtip vortex. This trend compares well with the experimental results shown in Figure 7.

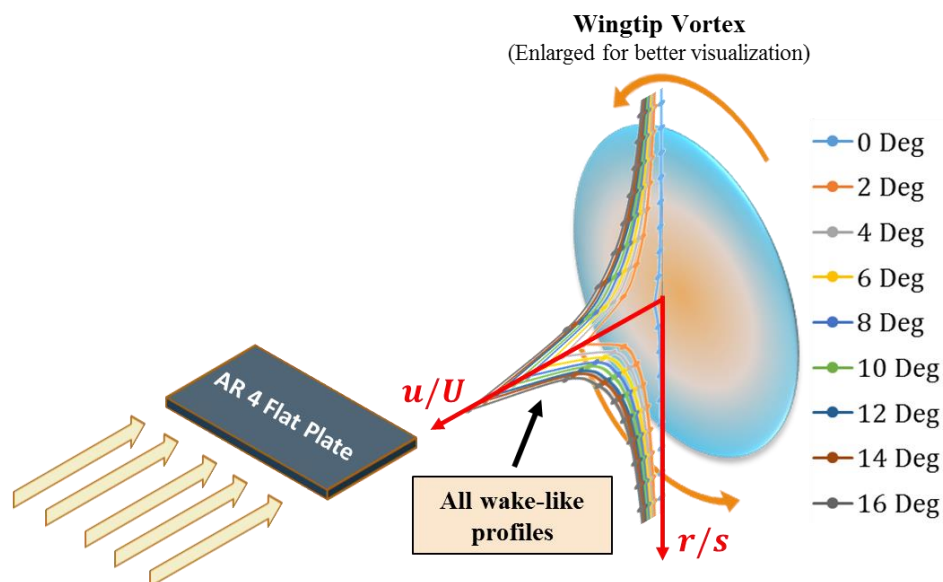


Figure 8 Prediction of wingtip vortex core axial velocity using Brown's model. The model shows a decrease in peak axial velocity of the wingtip vortex with increase in angle of attack. A similar trend is seen in the experimental results shown in Figure 7.

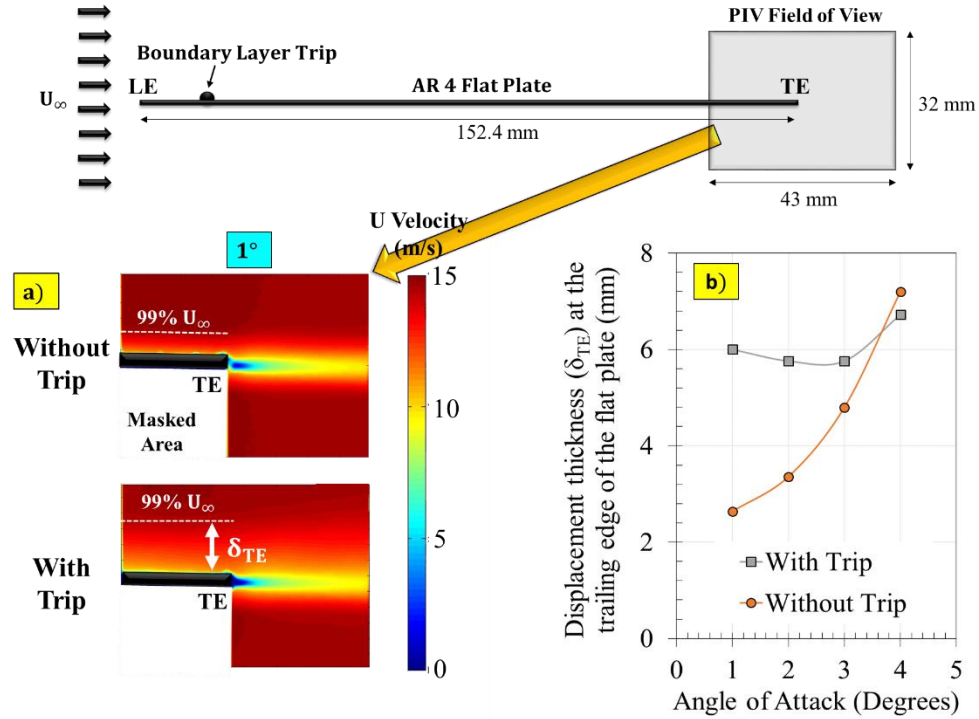


Figure 9 U Velocity contours of the trailing edge boundary layer of the flat plate a) with and without a BLT for 1° angle of attack b) The trailing edge displacement thickness is almost constant as a function of angle of attack in the tripped case whereas the trailing edge displacement thickness in the no-trip case increases with increase in the angle of attack.

Brown hypothesized that the decrease in the wingtip vortex peak core axial velocity is associated with the increase in the profile drag of the wing (Equation 7). Following Batchelor's model, the decrease in the wingtip vortex peak axial velocity is associated with the decrease in the wingtip vortex core radius and increase in azimuthal velocity. The boundary layer thickness increases with increase in angle of attack. This results in an increase in the skin friction drag of the wing which in turn increases the profile drag of the wing. The PIV results for the AR 4 flat plate trailing edge boundary layer are shown in Figure 9 for both the trip and the no-trip cases. As expected in the no-trip case, an increase in angle of attack increases the displacement thickness of the boundary layer. In contrast a relatively constant boundary layer thickness is seen in the tripped cases (Figure 9b) across this limited range of angle of attack. Since the coefficient of local skin friction and the normalized displacement thickness are inversely proportional to the $\sqrt{Re_c}$, where Re_c is the Reynolds number based on local chord length, then the local skin friction coefficient is proportional to the local displacement thickness,

$$C_{fx} \propto \frac{\delta^*}{x} \quad (13)$$

This relationship is applicable because there is no curvature (hence no pressure gradient) to the flat plate at 0° angle of attack. Since only lower angles of attack is considered in this study, any change in pressure gradient due to change in angle of attack can be assumed to be relatively small. Even though this assumption is not technically valid, it does provide a gross estimate of how the boundary layer is functionally related to the skin friction drag. The total skin friction drag of the wing increases with increase in the displacement thickness of the boundary layer. Hence the absolute skin friction drag is expected to increase with increase in angle of attack until large-scale separation and/or stall. At higher angles of attack, the pressure drag coefficient begins to dominate the drag component.

Correlation of Wingtip Vortex Results with Batchelor's Model

Batchelor's model indicates that the increase in wingtip vortex azimuthal velocity should result in a decrease in wingtip vortex axial velocity. An increase in the experimentally determined peak azimuthal velocity with an increase in angle of attack can be seen in Figure 10. As shown in Figure 7 and Figure 11, the wingtip vortex axial velocity decreases for 3° and 4° angle of attack. This is in good agreement with Batchelor's model. At 1° and 2° angle of attack little change in wingtip vortex axial velocity is observed.

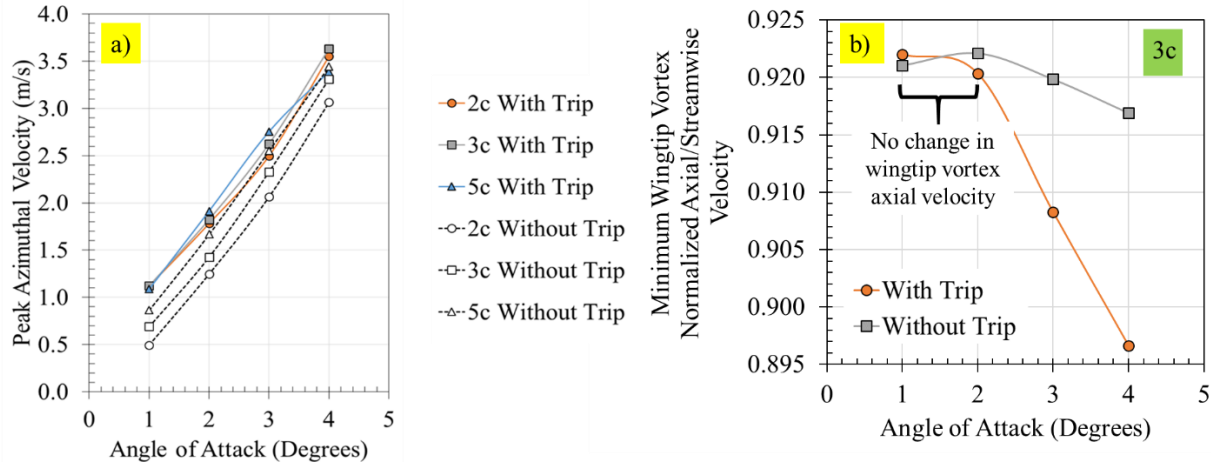


Figure 10 a) Increase in wingtip vortex peak azimuthal velocity with angle of attack b) Decrease in wingtip vortex minimum axial/streamwise velocity with angle of attack

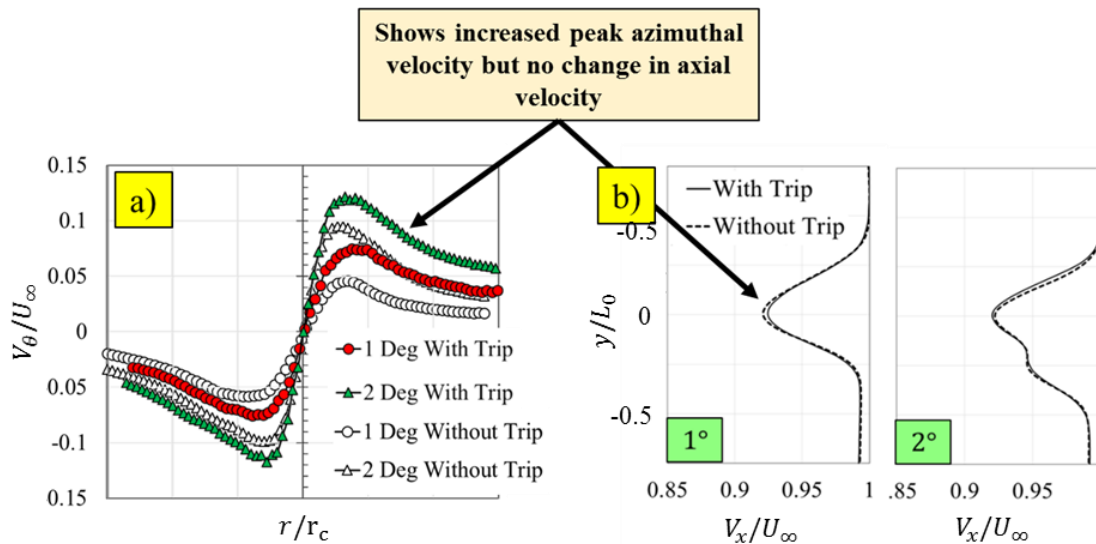


Figure 11 a) The BLT increased the peak azimuthal velocity when compared to the no-trip case. b) But no changes are observed in the axial/streamwise velocity in the tripped case at lower angles of attack. Hence in this case Batchelor's model does not conserve momentum.

The presence of the BLT increased the peak azimuthal velocity at all angles of attack (Figure 10a). While the increase in wingtip vortex azimuthal velocity in the tripped case resulted in a decrease in wingtip vortex axial velocity at 3° and 4° angles of attack, no discernable changes are seen in the axial velocity at 1° and 2° angles of attack (Figure 10b). The freestream normalized azimuthal velocity profile (Figure 11a) clearly indicates differences in the velocity magnitude between the trip and the no-trip case at 1° and 2°

angles of attack. However, the wingtip vortex axial velocity profile (Figure 11b) shows no corresponding variation between trip and the no-trip case at these angles of attack.

Therefore for the tripped case and the no-trip case, the 1° and 2° angles of attack seem to violate the underlying assumptions behind Batchelor's model. It is hypothesized that the presence of wingtip vortex-FSL interaction at lower angles of attack induces viscous effects which inhibit applicability of Batchelor's model at lower angles of attack. Also, it is evident that the wingtip vortex is not well-formed at lower angles of which may also inhibit the applicability of Batchelor's model at lower angles of attack. One manner in which to identify any shortcomings in Batchelor's model at lower angles of attack is through the application of conservation of momentum in the wingtip vortex. This topic is revisited in subsequent paragraphs.

Batchelor's model in Equation 5 is derived with an inviscid assumption. At lower angles of attack, the FSL interaction with the wingtip vortex is believed to amplify the effect of viscosity which might result in a violation of the underlying assumptions in Batchelor's model. Batchelor's model shown in Equation 5 is modeled with an assumption that the radius of the vortex and the total pressure in the vortex are not a function of viscosity. In reality, viscous forces increase the vortex radius and reduce the total pressure which affects the balance between the azimuthal velocity and axial velocity as described by the model. This functional relationship can be shown by using the momentum equation in the axial direction. Using Figure 4 for the system definition, the momentum equation in the axial direction including viscous effects can be written as,

$$V_x \frac{\partial V_x}{\partial x} + V_r \frac{\partial V_x}{\partial r} + \nu \nabla^2 V_x = -\frac{1}{\rho} \frac{\partial P}{\partial x} \quad (14)$$

Assuming minimal radial velocity in the wingtip vortex, Equation 14 reduces to

$$V_x \frac{\partial V_x}{\partial x} + \nu \nabla^2 V_x = -\frac{1}{\rho} \frac{\partial P}{\partial r} \frac{\partial r}{\partial x} \quad (15)$$

Since the relationship between pressure and azimuthal velocity of the vortex shown in Equation 2 applies for both viscous and inviscid flows, it can be substituted into Equation 15.

$$V_x \frac{\partial V_x}{\partial x} + \nu \nabla^2 V_x = -\frac{V_\theta^2}{r} \frac{\partial r}{\partial x} \quad (16)$$

Multiplying dx throughout Equation 16,

$$V_x \partial V_x + \nu \nabla^2 V_x \partial x = -\frac{V_\theta^2}{r} \partial r \quad (17)$$

Integrating Equation 17.

$$\frac{V_x^2}{2} + \nu \int \nabla^2 V_x \partial x = -V_\theta^2 \ln r \quad (18)$$

Dividing Equation 19 by U_∞^2 ,

$$\frac{V_x^2}{U_\infty^2} + \frac{\nu}{U_\infty^2} \int \nabla^2 V_x \partial x = -\left(\frac{V_\theta}{U_\infty}\right)^2 \ln r \quad (19)$$

From Equation 20, neglecting viscous forces, the following can be concluded:

- Assuming azimuthal velocity constant, an increase in vortex radius will result in a decrease in axial velocity which is in agreement with the results shown in Equation 5 for momentum in the radial direction.
- Assuming azimuthal velocity is increasing as a function of angle of attack, the wingtip vortex axial velocity should decrease which is accordance with Batchelor's model. The radius of the vortex will also increase as a function of angle of attack which will decrease the axial velocity as mentioned in the prior case.

If the viscous term is considered in Equation 20, the following can be concluded:

- The functional relationship mentioned in the inviscid cases between the wingtip vortex azimuthal velocity, radius and axial velocity will no longer be applicable when viscous forces are considered. This might be why there is no visible change in the wingtip vortex axial velocity at 1° and 2° angles of attack between the tripped and the no-trip cases (Figure 11b), even though there is an increase in azimuthal velocity and circulation in the tripped case when compared to the no-trip case. But as observed in Figure 10b, at 3° and 4° angles of attack, there is an apparent change in the axial velocity which might be due to the reduction of the influence of viscous effects caused by the separation of the free shear layer from the wingtip vortex.
- Comparing the results in Figure 11b and Equation 20, the axial velocity remains the same for the tripped and the no-trip case at 1° and 2° angles of attack. Without a well-formed wingtip vortex at lower angles of attack as seen in Figure 6, Figure 7 and Figure 11, an accurate estimation of the radius of the vortex boundary is not possible. However, by comparing the wingtip vortex axial velocity profile in Figure 11b, there is no change in the radius of the wingtip vortex contained shear layer between the trip and the no-trip case. It can thus be assumed that the radius of the vortex remains relatively constant at those angles of attack. Since the azimuthal velocity is increased for both cases, according to Equation 20, to conserve axial momentum, the magnitude of the viscous term has to increase to balance the equation.
- The viscous dissipation term in Equation 20 was determined numerically from experimental results and is shown in Figure 12 for 1° and 2° angle of attack for the trip and the no-trip cases. As Equation 20 suggests, the viscous dissipation for the trip cases is increased for both angles of attack to conserve momentum.

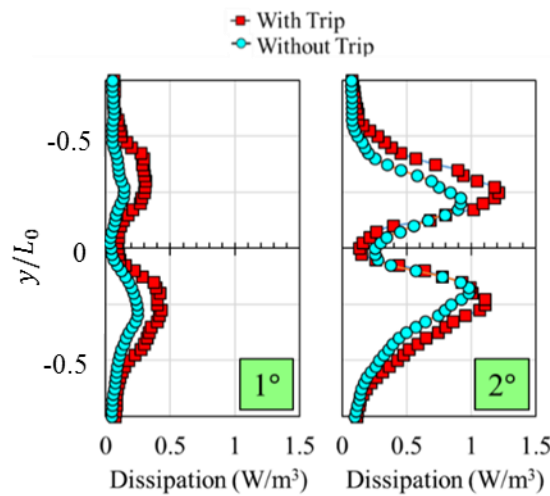


Figure 12 Represents the increases in viscous dissipation for the tripped case over the no trip case necessary to conserve momentum in the streamwise direction according to Equation 20.

Correlation of Wingtip Vortex Results with Brown's Model

Brown's model was based on the hypothesis that the profile drag of the wing is enveloped in the rollup of the wingtip vortex. From the trailing edge boundary layer results shown in Figure 9, using Equation 13, it can be stated that the profile drag of the wing increases with increase in angle of attack. With the increase in profile drag, there are clear changes in the wingtip vortex axial velocity as observed in Figure 11b at 3° and 4° angles of attack for both the trip and the no-trip case. However, at 1° and 2° angles of attack, no change in axial velocity is observed resulting from the increase in profile drag of the wing.

From Figure 9, at lower angles of attack, the displacement thickness at the trailing edge of the BLT wing is significantly larger than the displacement thickness in the no-trip case (Figure 9b). Therefore, the profile drag of the wing is increased by the BLT at lower angles of attack. In contrast, little to no difference is seen in the wingtip vortex core axial velocity between the trip and the no-trip case at lower angles of attack. It is evident that at lower angles of attack, the wingtip vortex is not fully evolved due to the wingtip vortex-FSL interaction (also shown in Figures 6, 7 and 11b). As the FSL bifurcates from the wingtip vortex, a wingtip vortex core is distinguished which has a strong axial current.

It is possible that Brown's model is not applicable at lower angles of attack due to the extent of the FSL interference with the wingtip vortex as mentioned in the previous section. One potential explanation for Brown's model not applying as well at lower angles of attack is that the magnitude of the spanwise flow over the wing is less when compared to higher angles of attack. Hence the boundary layer (profile drag) of the wing could not be fully enveloped in the roll-up process of the wingtip vortex as hypothesized by Brown.

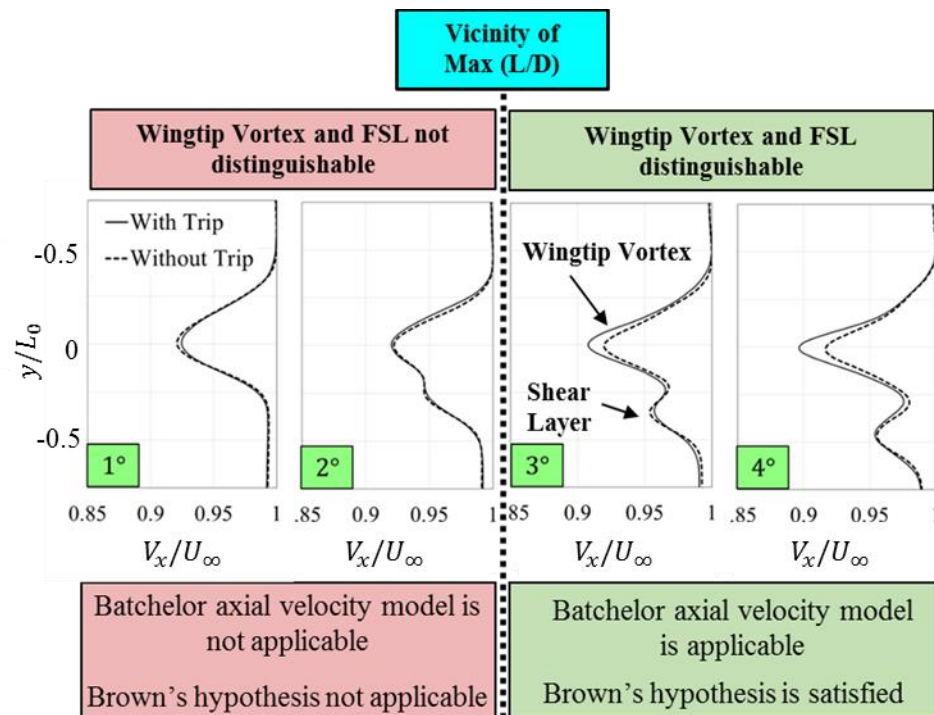


Figure 13 The streamwise/axial velocity profiles at the wingtip vortex station 3 chord lengths downstream indicate at lower angles of attack, the FSL and the wingtip vortex are not distinguishable. Batchelor's and Brown's models are not applicable at lower angles of attack.

A summary of the comparison of the wingtip vortex axial/streamwise velocity with Batchelor's and Brown's models is shown in Figure 13. A more clear definition of the wingtip vortex and FSL flowfield is

observed at 3° and 4° angle of attack around the vicinity of maximum (L/D) condition where Batchelor's model and Brown's hypothesis are more applicable. Consequently, when the angle of attack is increased, the wingtip vortex core axial velocity for the trip and the no-trip cases can be readily differentiated. The decreased core axial velocity in the tripped case (when compared to the no-trip case) could be due to the increase in profile drag of the wing following from Brown's model. It could also potentially be attributed to increased azimuthal velocity and added vorticity at higher angle of attack according to Batchelor's model. It is non-trivial to distinguish the contribution to the decrease in the wingtip vortex core axial velocity from either of the two.

The Batchelor model for the axial velocity is compared with the wingtip vortex axial velocity profiles at 3° and 4° angle of attack in Figure 14. A rescaling value (l) of 2 is used in Batchelor's model shown in Figure 14. This agrees well with the present experimental results, particularly at 3° angle of attack. This rescaling parameter is a function of peak azimuthal velocity of the wingtip vortex. Minor deviations are seen between the Batchelor's model and in the wingtip vortex inner core at 4° angle of attack. It can also be inferred from Figure 14 that the BLT did not affect the nature of the profile of the wingtip vortex inner core axial velocity but affected the magnitude of the peak axial velocity as shown in Figure 7 and Figure 10b.

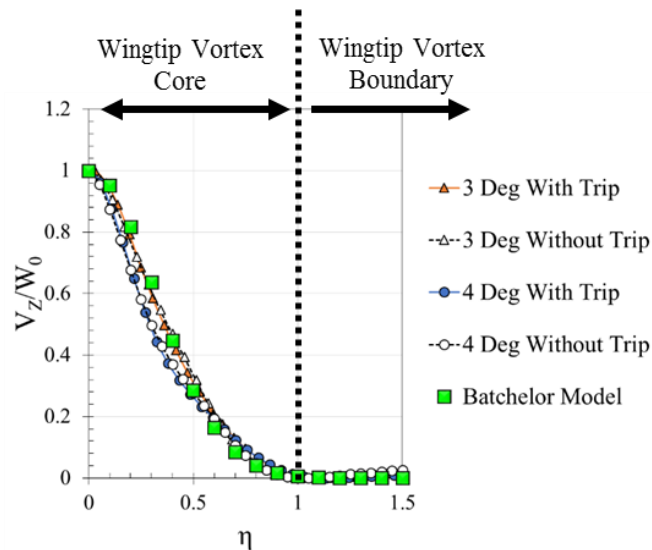


Figure 14 Batchelor axial velocity model agrees well with experimental data for 3° angle of attack. Minor variations are seen between the Batchelor's model and 4° angle of attack case. This difference might be due to the increase in peak azimuthal velocity as a function of angle of attack. The deviations in the axial velocity distributions lie within the uncertainty bounds of the PIV results.

4.2 Interaction Between the FSL and the Wingtip Vortex Azimuthal Velocity

Similar to the previous section, the peak normalized azimuthal velocity profiles were compared to the well-known Batchelor's azimuthal velocity model in order to get a deeper understanding of the relationship between the on-body boundary layer or the profile drag and the azimuthal velocity of the wingtip vortex.

The comparison between Batchelor's model and the experimentally derived wingtip vortex azimuthal velocity measured 2 chord lengths downstream is shown in Figure 15 for all angles of attack tested. In both the trip and the no-trip cases, the normalized azimuthal velocity profiles show good correlation with Batchelor's model at all angles of attack. At a 4° angle of attack, the experimental azimuthal profile and Batchelor's model agree comparatively well for the trip and the no-trip cases. However, the

departure from Batchelor's model can be observed at the wingtip vortex core boundary in the tripped case (Figure 15a) at lower angles of attack.

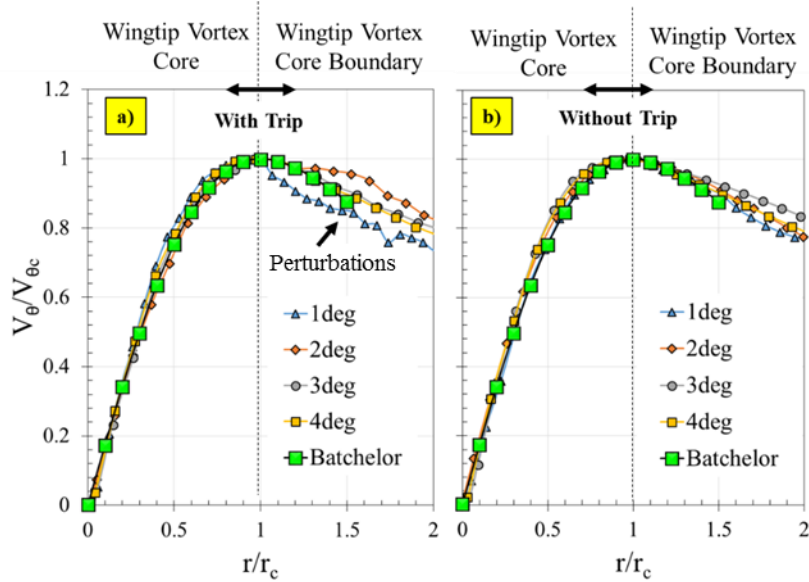


Figure 15 Comparison of the experimental wingtip vortex azimuthal velocity profiles with the Batchelor's model taken at 2 chord lengths downstream behind a) Tripped flat plate b) No-trip flat plate. The tripped case shows deviation from the Batchelor's model in the wingtip vortex core boundary at lower angles of attack.

To understand the changes in the wingtip vortex core outer boundary azimuthal profile at lower angles of attack, velocity fields derived from instantaneous PIV image pairs were used to create streamlines to visualize the roll-up process of the wingtip vortex. Representative cross-stream images of instantaneous streamlines of the wingtip vortex taken at 2 chord lengths downstream of the trailing edge of the wing are shown for all angles of attack considered in Figure 15. At a 1° angle of attack, the wingtip vortex is not fully formed and the roll-up of the wingtip vortex appears distorted. There are distinctly identifiable perturbations in the boundary of the wingtip vortex. These perturbations are reflected in the normalized azimuthal velocity profile shown in Figure 15. As the angle of attack is increased, roll-up of the wingtip vortex is more orderly. At a 4° angle of attack, the wingtip vortex is fully formed and the level of disruption along the wingtip vortex boundary is drastically reduced when compared to the 1° case. Another manner to view the azimuthal wingtip vortex rollup process is to evaluate the vortex Reynolds numbers as a function of angle of attack for various distances downstream in Figure 16.

The vortex Reynolds number in Figure 16a was calculated using the peak azimuthal velocity (V_θ) and the wingtip vortex outer core radius (r_{core}) as the characteristic length [8]. The wingtip vortex outer core is estimated from the asymptotic decrease in the azimuthal velocity profile where the changes in the azimuthal velocity in an interval of $0.2 r/r_c$ is less than 1%.

$$Re_{vortex} = \frac{\rho V_\theta r_{core}}{\mu} \quad (20)$$

where ρ is the density and μ is the dynamic viscosity. The vortex Reynolds number varies almost linearly with angle of attack for both the trip and the no-trip cases. This linear variation in the vortex Reynolds number might be due to the linear variation of vortex core radius and/or peak azimuthal velocity as shown in the previous study [1]. The variation in circulation in the wingtip vortex also showed linear behavior. This behavior would be expected as predicted from the Kutta-Joukowski theorem, the linearity associated with all of these variations is coupled to the linearity in the lift. For the no-trip case, the vortex Reynolds

number increases as the distance downstream increases but for the trip case, the vortex Reynolds numbers coalesce for different distances downstream. The percentage difference in the vortex Reynolds number between the trip and the no-trip cases (Figure 16b) decreases with increase in angle of attack. Similar behavior is seen in the mid-semispan FSL as well [1]. This indicates that at lower angles of attack the FSL content gets enveloped in the wingtip vortex core affecting the peak azimuthal velocity and the vortex core radius and hence the vortex Reynolds number. It is interesting to note that the BLT did not affect the axial core velocity profile at lower angles of attack and only effected the cross-stream azimuthal velocity profile of the wingtip vortex. With the increase in angle of attack, the FSL diverges from the wingtip vortex (Figure 5) which increases the peak azimuthal velocity and vortex core radius and therefore results in an increase in the vortex Reynolds number. The vortex core radius could also potentially increase due to the divergence of the tip vortex from the FSL, but it could also occur quite naturally as a simple function of increase in angle of attack. It is difficult to distinguish the contribution of the two factors.

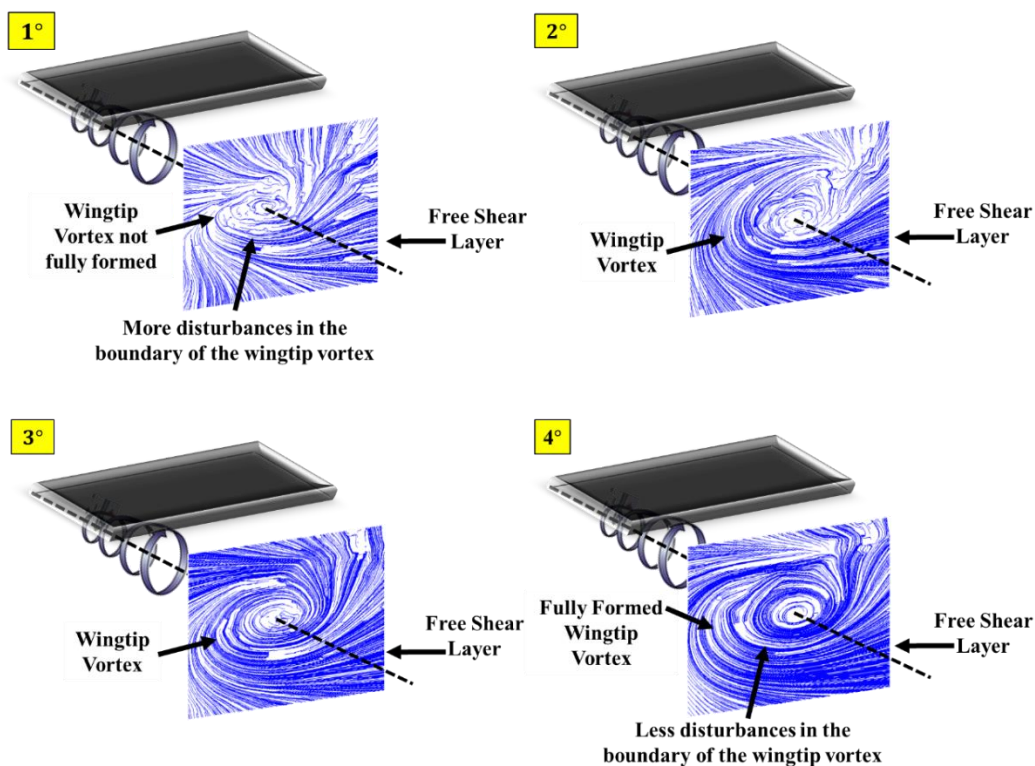


Figure 16 Instantaneous streamlines of the wingtip vortex taken 2 chord lengths downstream at several angles of attack. At a lower angle of attack, the wingtip vortex is not fully formed with greater perturbations seen in the wingtip vortex outer core boundary. At higher angles of attack, the roll-up of the wingtip vortex is less orderly in the wingtip vortex outer core boundary. This change in the nature of the azimuthal velocity distribution is reflected in the normalized profiles seen in Figure 15.

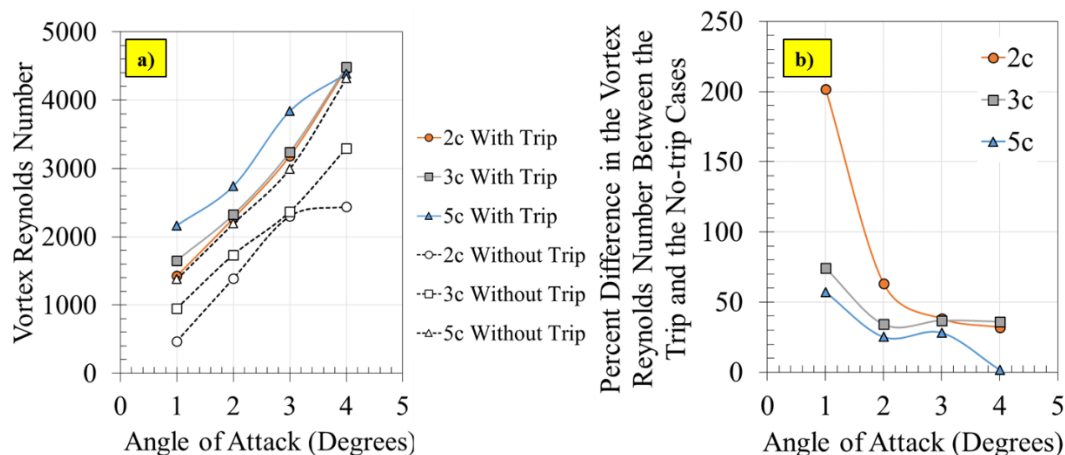


Figure 17 a) The vortex Reynolds number increases linearly with increase in angle of attack resembling the peak azimuthal velocity and circulation plotted as a function of angle of attack. b) The decreasing percent difference in the vortex Reynolds number between the trip and the no-trip cases shows similar behavior as the mid semispan FSL. This indicates at lower angles of attack, the mid semispan FSL content is rolled up in the wingtip vortex.

The disturbances observed in the azimuthal velocity profile and the deviation from the behavior of Batchelor's model at lower angles of attack is hypothesized to be due to the enhanced interaction between the FSL and the wingtip vortex resulting from the BLT. This hypothesis was substantiated by analyzing the wingtip vortex PIV results in the spanwise plane. The isometric view of the spanwise planes at different angles of attack and at different chord lengths downstream is shown in Figure 17. At a 1° angle of attack, the FSL and the wingtip vortex are seen in the same plane and in sufficient proximity to have a significant interaction. This is especially true for the near-wake distances. As the angle of attack increases, the FSL deviates below the plane of the wingtip vortex. At a 2° angle of attack, the FSL is only distinguishable 2 chord lengths downstream. At 3° angle of attack, the FSL moves out of the plane of the wingtip vortex entirely. As the distance between the wingtip vortex and the shear layer increases, the interaction between them decreases. This bifurcation of the wingtip vortex and the FSL was shown earlier in the streamwise plane in Figure 6. It is remarkable that this occurs in the vicinity of the maximum (L/D) lift condition.

The significant increment in profile drag induced by the BLT at lower angles of attack not only increased the magnitude of the peak azimuthal velocity but also changed the nature of the vortex core boundary due to the FSL interaction. Brown hypothesized that one way to model the vortex was to assume all of the profile drag of the wing is enveloped in the wingtip vortex core during the rollup process. The results in Figure 11 and Figure 17 indicate that the change in profile drag also changes the nature of the azimuthal velocity distribution of the wingtip vortex in the near wake. The comparison of the normalized peak azimuthal velocity variation with downstream distance is shown in Figure 18 for a 1° angle of attack. The 2 chord lengths downstream case shows significant deviation from Batchelor's model in the wingtip vortex core boundary due to the proximity of the FSL as shown in Figure 15. As the downstream distance increases, the interaction between the FSL and the wingtip vortex is reduced. This explanation is corroborated by the good match between Batchelor's model and the normalized peak azimuthal velocity at 3 and 5 chord lengths downstream in Figure 19.

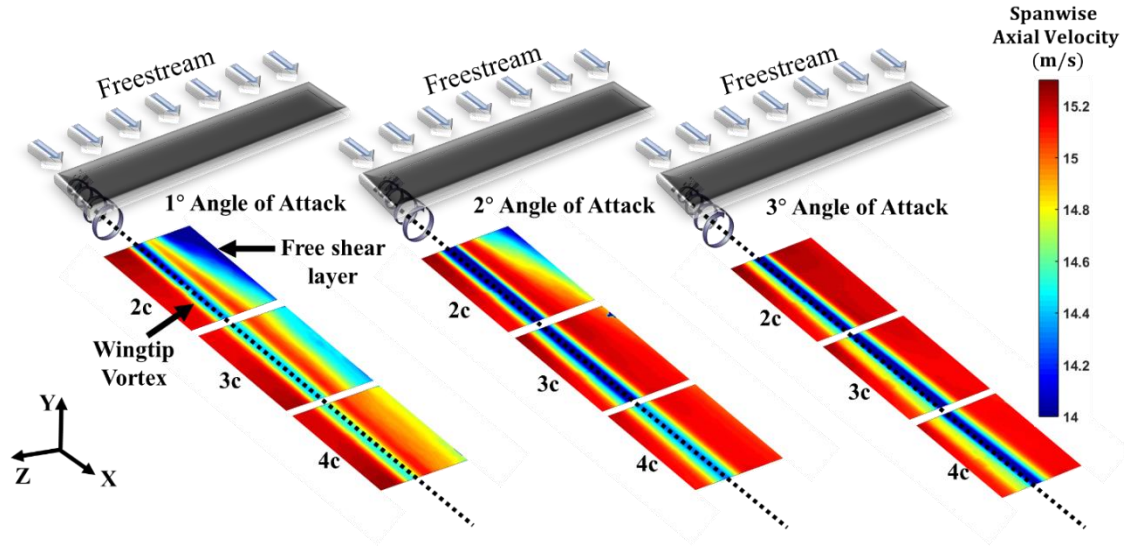


Figure 18 Spanwise PIV contours of the wingtip vortex at several chord lengths downstream and at several angles of attack. The proximity of the FSL to the wingtip vortex at lower angles of attack ($2c$, 1°) is significant enough to affect the rollup of the wingtip vortex as seen in the normalized azimuthal velocity profiles shown in Figure 15.

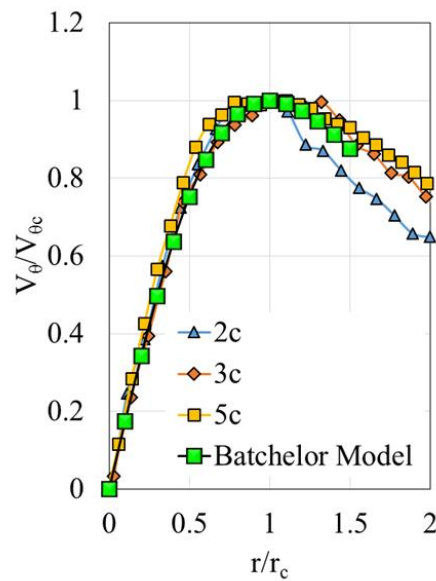


Figure 19 Comparison of Batchelor's model with normalized peak velocity of the wingtip vortex of the flat plate at a 1° angle of attack. The proximity of the free shear layer at shorter downstream distances changes the nature of the azimuthal velocity distribution of the wingtip vortex.

4.3 Cross-over of Momentum from the FSL to the Wingtip Vortex

Gunasekaran and Altman [1] showed that the addition of the BLT changed the nature of the boundary layer which in turn influenced the nature of the FSL and the wingtip vortex azimuthal velocity at lower angles of attack where skin friction drag is the dominating factor. At higher angles of attack, the differences between the trip and no trip case in the trailing edge displacement thickness decreased. This was reflected in the mid semispan FSL (Figure 20a) and the azimuthal velocity and circulation of the wingtip vortex [1]. The larger change in the skin friction drag was not reflected in the wingtip vortex axial

core velocity at lower angles of attack (Figure 20b), however, since the percentage difference in the peak core axial velocity between the trip and no trip case is at a minimum at lower angles of attack.

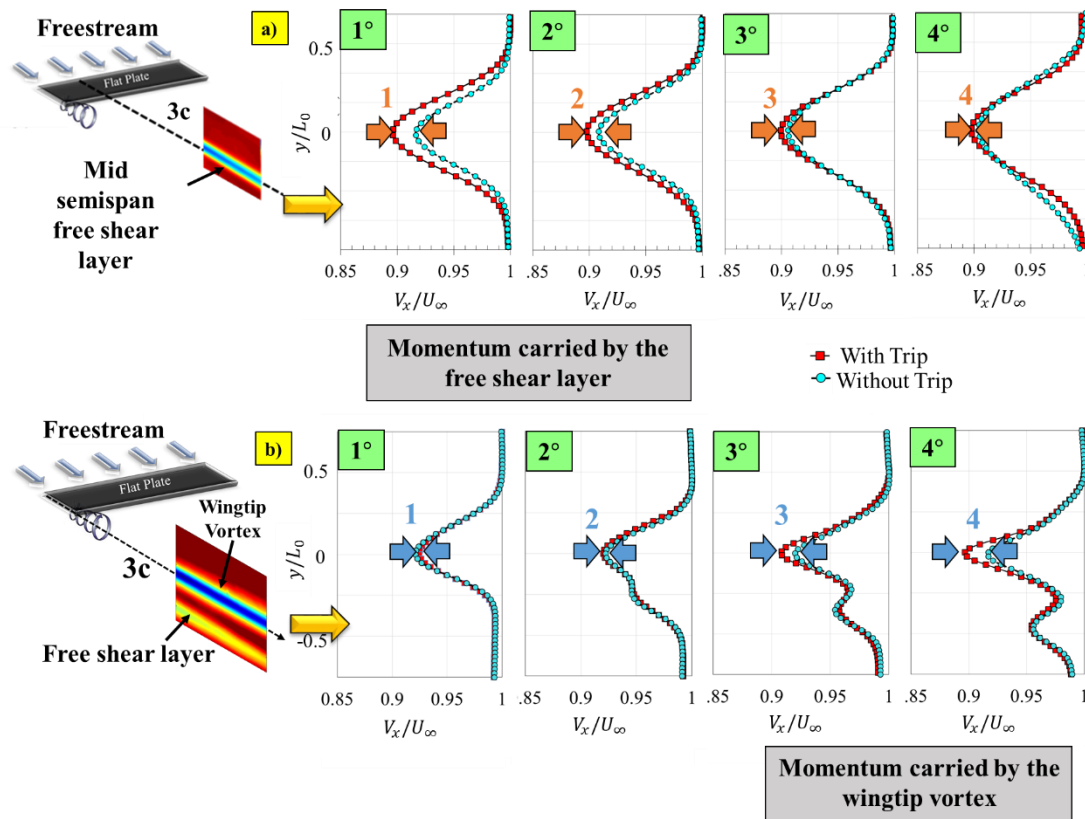


Figure 20 At lower angles of attack the momentum is carried by the FSL (a). At higher angles of attack the momentum is carried by the wingtip vortex (b). Similar results are observed at all distances downstream tested.

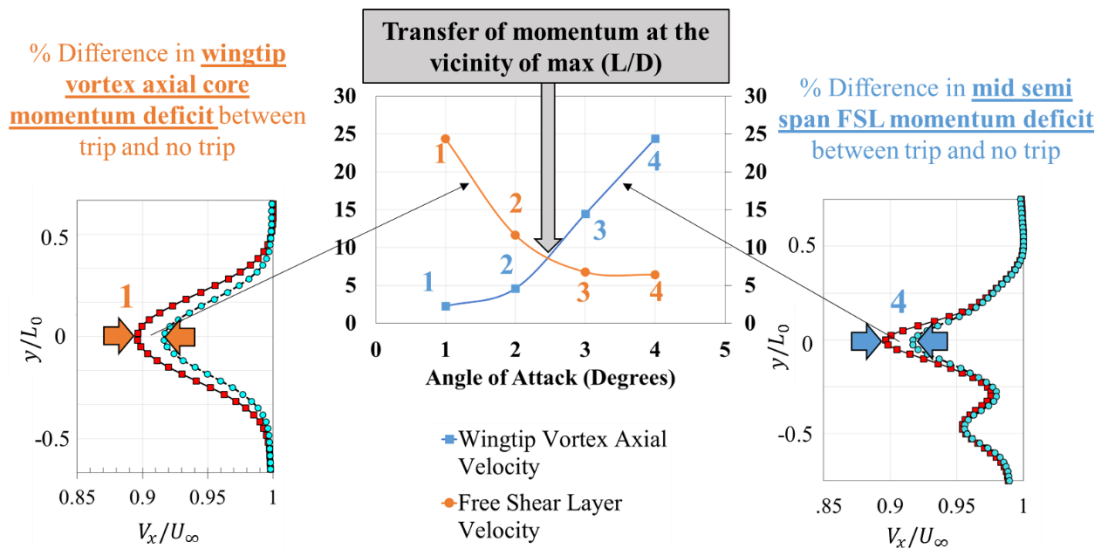


Figure 21 The cross-over of the momentum shown in Figure 20 occurs in the vicinity of the maximum (L/D) lift condition. Similar results are observed at all distances downstream tested.

The results shown in Figure 17b suggest that the changes in the boundary layer due to the BLT go into the roll up of the wingtip vortex and not into the FSL at 3° and 4° angle of attack. The loss of freestream momentum due to the BLT is transferred from the FSL to the wingtip vortex as the angle of attack is increased. This transfer of the loss of freestream momentum occurs in the vicinity of the maximum (L/D) lift condition as shown in Figure 21 and is hypothesized to occur in the near wake due to the pressure imbalance imposed by the tip vortex.

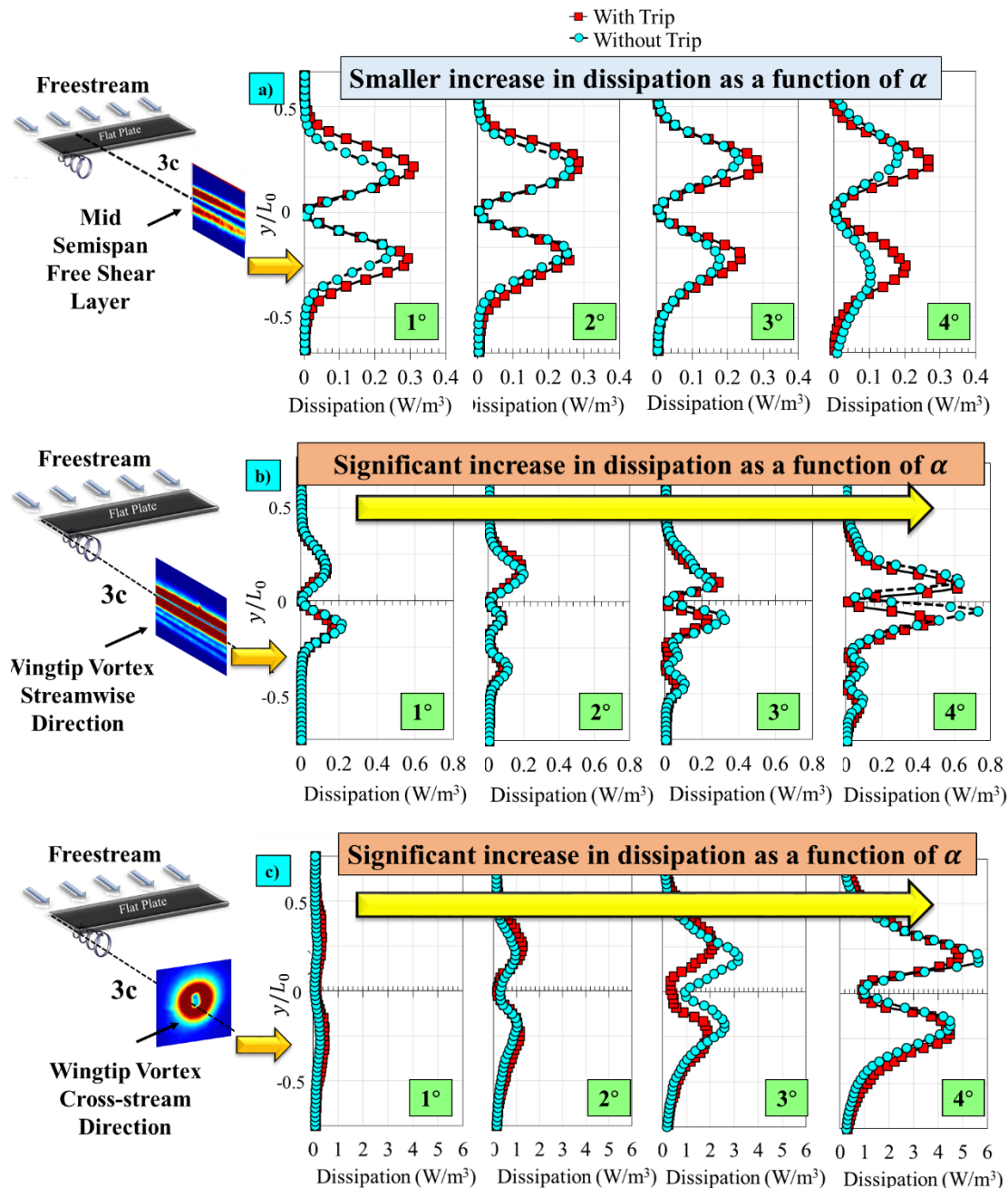


Figure 22 Variation of viscous dissipation in a) Mid semispan FSL b) Streamwise wingtip vortex c) Cross-stream wingtip vortex. A significant increase in the viscous dissipation in the wingtip vortex with increase in angle of attack indicates more losses in the wingtip vortex when compared to the FSL.

In the freestream normalized mid semispan FSL velocity profiles the differences between trip and no trip case start to decrease at an angle of attack of 3° (Figure 20a). The differences in the wingtip vortex axial velocity profiles between the trip and no trip case increase at an angle of attack of 3° (Figure 20b). The comparison of the momentum deficit in the mid semispan FSL velocity profile and wingtip vortex core axial velocity profile show a cross-over of the momentum between 2° and 3° angles of attack (Figure 20c). Between these angles of attack the FSL bifurcates from the wingtip vortex as observed in the streamwise axial velocity contour of the wingtip vortex shown in Figure 6. The correlation of this variation with the location of maximum (L/D) lift condition is noteworthy and is found at all downstream distances tested. This behavior and interaction between the wingtip vortex and shear layer could potentially evince serious implications on increasing aerodynamic efficiency.

The increase in drag (as represented by an isolated element of the momentum wake deficit drag formulation) at the wingtip vortex at higher angles of attack can be supported by considering the viscous dissipation of the cross-stream wingtip vortex in Figure 22. Since the viscous dissipation is one way to represent losses in the flowfield, assuming zero thermal conduction, zero body force and constant internal energy, changes in the viscous dissipation can act as a surrogate for changes in aerodynamic drag. The viscous dissipation of the mid semispan FSL shows minuscule variation with increase in angle of attack for both the trip and the no-trip case (Figure 22a). A significant increase in viscous dissipation, however, is observed in the wingtip vortex with increase in angle of attack both in the streamwise direction (Figure 22b) and in the cross-stream direction (Figure 22c). Since the velocity gradients contributing to circulation also contribute to viscous dissipation, the increase in the viscous dissipation with increase in angle of attack in the cross-stream direction is in agreement with expectation. No such relationship is evident between the streamwise viscous dissipation and the circulation. However, the magnitude of the increase in streamwise and cross-stream viscous dissipation implies that there is an interaction between the two axes as they demonstrate highly similar behavior.

The viscous dissipation in the cross-stream direction in the mid semispan FSL is assumed to be small when compared to the dissipation in the freestream aligned direction. Therefore the increase in viscous dissipation in the wingtip vortex in both the streamwise and the cross-stream direction represents an increase in losses as observed in the streamwise momentum deficit in Figure 20.

5. Conclusion

This study provided clear evidence of the interaction between the FSL and the wingtip vortex and their collective relationship to the aerodynamic efficiency.

- At lower angles of attack, the wingtip vortex core boundary deviates from Batchelor's azimuthal velocity model. This shows that Batchelor's model is less applicable at lower angles of attack where an interaction between the FSL and the wingtip vortex are more pronounced.
- Brown's model based on the concept of the profile drag being enveloped in the rollup of the wingtip vortex also saw limited applicability at lower angles of attack.
- The FSL deviates from the wingtip vortex in the vicinity of maximum (L/D) resulting in an easily distinguishable wingtip vortex and reduced FSL interaction with the wingtip vortex. Therefore, in the cross-stream direction at angles of attack lower than maximum (L/D), the FSL interaction with the wingtip vortex is significant. At angles of attack higher than those corresponding to maximum (L/D), the interaction attenuates.

If an airplane wing were to be modified to influence when the FSL deviates from the wingtip vortex, it is believed that the aerodynamic efficiency can be improved in off-design conditions.

- At angles of attack higher than maximum (L/D), the peak axial core velocity shows greater deficit in the tripped case when compared to the peak axial core velocity at lower angles of attack. This increase

in the “drag” of the wingtip vortex is due to the transfer of momentum from the FSL to the wingtip vortex. This conclusion was derived by determining the differences between the trip and the no-trip case in the wingtip vortex axial velocity and in the mid semispan FSL.

- At lower angles of attack, the additional loss of momentum due to the BLT is manifested in the FSL. At higher angles of attack, the additional loss of momentum due to the boundary layer trip is contained within by the wingtip vortex. This cross-over of momentum happens in the vicinity of maximum (L/D).

It is thus believed that the aerodynamic efficiency can potentially be improved in off-design conditions by manipulating the cross-stream momentum leaving the trailing edge of the wing of an airplane.

6. References

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